MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

COURSE PLAN **AY 2025-26 (Odd semester)**

Department:	**Electronics and Communication**	
Course Name & code:	**Network Analysis**	**ECE2122**
Semester & branch:	**IIIrd Semester**	**Electronics and Communication, Electronics in VLSI Design and Technology**
Name of the faculty:	**1. Dr. Somashekara Bhat** **2. Dr. R. Vinoth** **3. Dr. Jagadeesh Chandra R B** **4. Dr. Om Prakash Kumar** **5. Dr. Praveen Kumar**	

No of contact hours/week:	**L**	**T**	**P**	**C**
	3	**0**	**0**	**3**

COURSE OUTCOMES (COS)

At the end of this course, the student should be able to:		No. of Contact Hours	Marks	Program Outcomes (POs)	PSO	BL
CO1	*Apply mathematical techniques, such as loop analysis, nodal analysis, and network theorems, to analyze electrical networks and predict the system response to different stimuli.*	8	25	1,2,5,12	1	2
CO2	*Apply the knowledge of initial and final conditions in passive circuits to obtain the transient and steady-state responses of RL, RC, and RLC circuits, and demonstrate the ability to solve complex problems related to passive circuits.*	12	32	1,2,5,12	1	2,3
CO3	Evaluate the response of an RC circuit to a range of input signals, including step, pulse, square, and ramp inputs, and analyze the effect of different input signals on the behaviour of the circuit.	5	12	1,2,5,12	1	2
CO4	Utilize Laplace transform techniques to find the response of various electrical circuits.	5	15	1,2,5,12	1	2
CO5	Analyze two port networks to obtain their two port parameters, driving point functions, and transfer functions.	6	16	1,2,5,12	1	2
	Total	**36**	**100**			

COURSE LEARNING OUTCOMES (CLOS)

At the end of this course, the student should be able to:		No. of Contact Hours	Marks	Program Outcomes (POs)	Learning Outcomes (LOs)	BL
CLO1	*Apply mathematical techniques, such as loop analysis, nodal analysis, and network theorems, to analyze electrical networks and predict the system response to different stimuli*	8	25	1,2,5,12	C1	2
CLO2	*Apply the knowledge of initial and final conditions in passive circuits to obtain the transient and steady state responses of RL, RC, and RLC circuits, and demonstrate the ability to solve complex problems related to passive circuits.*	12	32	1,2,5,12	C3	2,3
CLO3	Evaluate the response of an RC circuit to a range of input signals, including step, pulse, square, and ramp inputs, and analyze the effect of different input signals on the behaviour of the circuit.	5	12	1,2,5,12	C2	2
CLO4	Utilize Laplace transform techniques to find the response of various electrical circuits.	5	15	1,2,5,12	C2	2
CLO5	Analyze two-port networks to obtain their two-port parameters, driving point functions, and transfer functions.	6	16	1,2,5,12	C2	2
	Total	**36**	**100**			

Assessment Plan

IN-SEMESTER ASSESSMENTS								
S. No.	**Assessment Mode**	**Assessment Method**	**Time Duration**	**Marks**	**Weightage**	**Typology of Questions (Recommended)**	**Schedule**	**Topics Covered**
1	**MISAC 1**	**Quiz/MCQs**	**20 Mins**	**5**	10 MCQs × ½ = 5 marks	Bloom's taxonomy (B) level of the question should be L2and above.		L1 – L8
	MISAC 2	**Mid-Term Examination**	**30 Mins**	**30**	Descriptive: 30 M	Bloom's taxonomy (BT) level of the question should be L2 and above.		L1 – L20
	MISAC 3	**Quiz/MCQs/ Surprise Assignment**	**15 Mins**	**5**	10 MCQs × ½ = 5 marks	Bloom's taxonomy (BT) level of the question should be L2 and above.		L21 – L25
	FISAC	**Quiz/MCQs/ Surprise Assignment/**	**-**	**10**		Bloom's taxonomy (BT) level of the question should be L3 and above.		L26- L36
END – SEMESTER ASSESSMENT								
1	**Regular/Make–Up Exam**		180 Mins	50	Answer all 5 full questions of 10 marks each. Each question can have 3 parts of 2/3/4/5/6 marks.	Bloom's taxonomy (BT) level of the question should be L3 and above.	17th week of the semester	Comprehensive examination covering full syllabus.

LESSON PLAN

L No	TOPICS	Course Outcome Addressed
1	Introduction -- Energy Sources-Voltage Source, Current source**.** Source Transformation, Kirchhoff's Laws, Star Delta Conversion	1
2	Analysis of networks containing dependent sources by node method of analysis.	1
3	Analysis of networks containing dependent sources by mesh method of analysis.	1
4	Network Theorems- Introduction, Superposition Theorem	1
5	Thevenin's Theorem	1
6	Norton's Theorem	1
7	Maximum Power Transfer Theorem	1
8	Reciprocity Theorem, Additional problems.	1
9	**Transient Analysis:** Introduction, The behaviour of circuit elements under switching conditions, and their representation	1
10	**First Order Circuits:** Evaluation of initial and final conditions in RL, RC for DC excitations	1
11	Evaluation of initial and final conditions in RLC Circuits for DC Excitations	2
12	Analysis of source free RL and RC circuits	2
13	Step response of RL and RC circuits - natural and forced response	2
14	**Second Order Circuits:** Evaluation of initial and final conditions in RLC circuits for DC excitations	2
15	Initial conditions in RLC circuit -contd	2
16	Analysis of source free RLC circuit	2
17	Analysis of source free RLC circuit- contd	2
18	Step response of series RLC circuit- natural and forced response	2
19	Step response of parallel RLC circuit- natural and forced response	2
20	General second order circuits	2
21	**Circuit Analysis in the s-domain**: Review of Laplace Transform: Definition, Transform for standard time functions, Properties, Inverse Laplace Transform,	4
22	Representation of R, L and C in s domain, Node band mesh analysis in the s-domain.	4
23	Response of RL, RC, and RLC networks using the Laplace Transform method.	4
24	Using Network theorems to analyze networks in S-domain.	4
25	Evaluating initial and final conditions using the initial and final theorems.	4
26	**Linear wave shaping:** Introduction, **RC** high pass circuit- Frequency response, step response, pulse response.	3
27	Response of High pass RC circuit to a square wave input	3
28	Response of High pass RC circuit to ramp input, RC circuit as differentiator, Example problems	3
29	**RC** low pass circuit- Frequency response, step response, pulse response.	3
30	Response of Low pass RC circuit to a square and ramp input, RC circuit as Integrator	3
31	**Network functions:** Introduction, Driving point impedance and admittance functions, transfer functions. Poles and Zeros of Network Functions,	5
32	Examples of finding the network functions.	5
33	Open circuit impedance parameters,	5
34	short circuit admittance parameters	5

35	Transmission parameters, Hybrid parameters.	5
36	Relationship between two port parameters, Parallel, Series & Cascade connection of two port networks	5

Course Articulation Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	2	2			2							2	2		
CO2	2	2			2							2	2		
CO3	3	2			2							2	2		
CO4	3	2			2							2	2		
CO5	3	2			2							2	2		
Articulation Level	2.6	2			2							2	2		

FACULTY MEMBERS TEACHING THE COURSE (IF MULTIPLE SECTIONS EXIST):

FACULTY	SECTION	FACULTY	SECTION
Dr. R. Vinoth	**A**	**Dr. Jagadeesh Chandra R B**	**D**
Dr. Praveen Kumar	**B**	**Dr. Somashekara Bhat**	**VLSI**
Dr. Om Prakash Kumar	**C**		

References:

1. M. E. Van Valkenberg, “Network analysis”, Prentice Hall of India, 2000.
2. William H. Hayt, Jack E. Kemmerly, Steven M Durbin, “Engineering Circuit Analysis”, 8th edition, Tata McGraw Hill India,
3. Millman, H. Taub, “Pulse, digital and switching waveforms”, 3nd Edition, *McGraw* Hill.
4. Ravish R Singh, “Network analysis and Synthesis”, McGraw Hill, 2013.
5. Joseph Edminister, “Electric Circuits”, Schaum’s Series, McGraw Hill.

* https://nptel.ac.in/courses/108102042
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Submitted by: Praveen kumar

(Signature of the faculty)

Date: 02-08-2025

Approved by:

(Signature of HOD)